



The research notes column reviews recent research studies in the field of strength and conditioning and provides brief applications to these studies.

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Current Research: Training for Strength, Power, and Speed

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SUMMARY

THIS RESEARCH BRIEF COVERS CONCURRENT STRENGTH AND ENDURANCE TRAINING IN ELITE SOCCER PLAYERS, THE EFFECT OF HYPERTROPHY TRAINING PROTOCOLS WITH MAXIMAL LOADING ON NEURAL DRIVE, THE EFFECT OF CLUSTER SETS ON POWER IN JUMP SQUATS, AND THE EFFECT OF AGE ON REPEATED-SPRINT ABILITY IN YOUTH SOCCER PLAYERS.

CONCURRENT STRENGTH AND ENDURANCE TRAINING IMPROVES MARKERS OF PERFORMANCE IN ELITE SOCCER PLAYERS

It is well established that strength and endurance are codeterminants for soccer-specific performance tasks. Specifically, maximal strength is often related to acceleration, overall movement velocity, and jumping ability. These types of strength power activities contribute to the most decisive moments in a soccer match, and the ability to repetitively perform these types of activities is of particular

importance to all levels of soccer players. However, much of the research on the relationship of strength, power, and high-intensity endurance capacity has been conducted on lower tier soccer players. To address this deficiency in the scientific literature, researchers from the Norwegian University of Science and Technology examined the effects of concurrent strength and endurance training on performance markers in elite soccer players. Specifically, 21 male, first league, elite soccer players between the ages of 20 and 31 years who recently took part in the Union des Associations Européennes de Football Champions' League participated in an 8-week training period that used aerobic interval training and high-intensity squat training. A pre-post research design was employed for this investigation because of the restrictions of working with elite athletes and the inability to find a suitable control population.

During this preseason period, a typical week of soccer training contained six 1.5-hour practice sessions plus a match that was used as a training tool was performed. Aerobic interval training consisting of 4 sets of 4-minute long

runs on a treadmill at a 5.5% incline at an intensity of 90–95% of heart rate max was performed 2 days per week. Between each aerobic interval, there was a 3-minute active recovery period performed at an intensity of 50–60% of heart rate max. After a 15-minute break, the subjects performed a high-intensity strength training intervention consisting of half squats performed to a 90° knee joint angle at an intensity equivalent to a 4-repetition maximum load. The load was modified to ensure 4 repetitions were performed during each of the 4 sets that comprised the training session. Three-minute recovery was given between each set. Several variables were assessed before and after the 8-week training period, including maximal squat strength, countermovement jump, and sprint performance. Additionally, maximal oxygen consumption ($\dot{V}O_{2\max}$), maximal heart rate, and running economy at 11 km·h⁻¹ were assessed.

Overall $\dot{V}O_{2\max}$ increased from 60.5 (51.1–67.1) mL·kg⁻¹·min⁻¹ to 65.7 (58.0–74.5) mL·kg⁻¹·min⁻¹ corresponding to a significant 8.6% (1.7–16.6, $p < 0.001$) increase. Maximal strength increased by 51.7% (13.3–135.3%) as

indicated by the increase in 1-repetition maximum back squat from 116 (85–150) to 176 (160–210) kg. Additionally, 10-m sprint time improved by 0.06 (0.02–0.16) seconds, whereas vertical jump increased by 3.0 (0.1–6.2) cm. These findings are particularly important as the athletes were engaged in a preseason training plan that consisted of a total 12.5 total hours of training per week, which consisted of 2.5 hours of warm-up, 1.5 hours of stretching, 1.5 hours of endurance running, 2.5 hours of small-sided games, 2.0 hours of technical training, 1.0 hour of strength training, and 1.5 hours of match play. Although this is an uncontrolled study, it is very valuable as it reveals that elite soccer teams can achieve strength and endurance gains in response to an appropriately structured training intervention.

Helgerud J, Rodas G, Kemi OJ, and Hoff J. Strength and endurance in elite football players. *Int J Sports Med* 32: 677–682, 2011.

HYPERTROPHY TRAINING WITH MAXIMUM LOADING PROTOCOLS RESULT IN REDUCTIONS IN NEURAL DRIVE TO THE MUSCLE

The acute fatigue of various resistance training loading paradigms is of particular interest to strength and conditioning professionals and sport scientists. Acute fatigue results in changes in force production capacities and can occur centrally or peripherally within the neuromuscular system. The location of where fatigue manifests has been linked to specific loading conditions that would suggest that maximal strength and hypertrophic protocols result in different locations of fatigue manifestation. To address this question, researchers from the Department of Biology of Physical Activity at the University of Jyväskylä in Finland examined the acute neuromuscular fatigue effects that occur in response to maximal strength and hypertrophy loading paradigms. Specifically, 13 untrained men undertook 2 experimental conditions that were separated by 1 week. The maximal strength protocol consisted of 15

repetitions performed at a 1-repetition maximum load, whereas the hypertrophy protocol consisted of 5 sets of 10 at a 10-repetition maximum load. In essence, this study examines the effects of repetition maximum training or training to failure on the manifestation of fatigue.

During each protocol, concentric load and muscle activity, electromyographic (EMG) amplitude, and muscle activity data were collected for each set. Additionally, maximal bilateral isometric force and muscle activation and blood lactate levels were assessed pre, mid, and up to 30 minutes post loading. During the maximum strength protocol, the concentric load was decreased after set 10, whereas no alterations in the load were made during the hypertrophy protocol. The hypertrophic protocol ($-48 \pm 9.7\%$) resulted in a significantly greater reduction in isometric force when compared with the maximal strength protocol ($-30 \pm 6.4\%$). Careful inspection of the responses to the hypertrophy training intervention revealed that there were large reductions in neuromuscular efficiency, and median frequency, while EMG amplitude was maintained. Additionally, this protocol resulted in significantly higher blood lactate levels.

Taken collectively, the authors concluded that the data indicated that the hypertrophy loading protocols result in significant peripheral fatigue. During the maximal strength protocol, evidence also suggested that peripheral fatigue had occurred. However, the magnitude of the markers of peripheral fatigue was much lower than that seen in the hypertrophy protocol. It is likely that different factors contributed to the peripheral fatigue associated with the maximal strength protocol. The data presented suggest that a reduced neural drive during the maximal strength intervention was the primary cause of fatigue. Overall, the researchers concluded that a complex interaction between peripheral and central fatigue factors contribute to the manifestation of fatigue during hypertrophy and maximal strength loading paradigms. Based on this study, it is also evident that

hypertrophic protocols create greater fatigue as indicated by the greater decrease in force production, increased lactate accumulation, and greater reductions in neuromuscular efficiency.

Walker S, Davis L, Avela J, and Häkkinen K. Neuromuscular fatigue during dynamic maximal strength and hypertrophic resistance loadings. *J Electromyogr Kinesiol* 2012. [Epub ahead of print January 13, 2012]

CLUSTER SETS ATTENUATE DECREASES IN POWER AND VELOCITY OF MOVEMENT DURING JUMP SQUAT TRAINING

There are numerous methods for introducing training variation into a strength training program, including manipulating sets, repetitions, load, exercise selection, and rest intervals. One additional method that can be considered is cluster or interrepetition rest training where the set is broken into small clusters of repetitions that alter the training stimulus provided by the set. Specifically, this method attempts to improve the force, velocity, and power profile of the training set. Although there are some scientific data that demonstrate the ability of the cluster set to improve these factors, there are limited data on ballistic movements such as the jump squat. To address the lack of research on ballistic movements, researchers from Edith Cowan University in Joondalup, Australia, recruited 20 male professional and semiprofessional rugby union players to participate in a study that examined the kinematics and kinetics of 3 different cluster models.

The study was a crossover design in which the subjects performed 4 sets of 6 repetitions of the jump squat with an absolute load of 40 kg with 4 different set structures. In a randomized order, each subject performed a training session using a traditional set structure and 3 different cluster set interventions. Cluster set model 1 required the subjects to perform 4 sets of 6 repetitions with 12-second rest between each repetitions. During cluster set model 2, subjects performed 4 sets of 6 repetitions

that were broken into doubles separated by 30 seconds. Finally, cluster set model 3 required 4 sets of 6 repetitions in which the set was broken into 2 clusters of 3 separated by 60 seconds. All kinematic and kinetic variables collected during the jump squat were averaged across the 4 sets. The biggest findings were that power and movement velocity decreased significantly in the latter repetitions of the traditional set, although the cluster set models employed were able to attenuate this reduction. Based on these findings, the authors suggested that the cluster set is able to maintain the quality of the training set and maybe a useful tool when attempting to maximize power and velocity development when using jump squat training.

Hansen KT, Cronin JB, and Newton MJ. The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *Int J Sports Physiol Perform* 6: 455–468, 2011.

DO FITNESS DETERMINANTS OF REPEATED-SPRINT ABILITY VARY WITH AGE IN YOUTH SOCCER PLAYERS?

There is a plethora of data available looking at the importance of repeated-sprint ability in soccer players. In highly trained youth populations, there has been some exploration into the

relationships between repeated-sprint ability and acceleration and maximum speed. However, there are limited data exploring the relationship between repeated-sprint ability and other fundamental fitness qualities such as agility, explosive leg power, and aerobic conditioning in youth populations. To address the lack of scientific inquiry exploring these relationships, Spencer et al. recently examined repeated sprints time (s), 15-m sprint velocity (m/s), 15-m agility speed (m/s) countermovement jump height (cm), and aerobic power as estimated from a Yo-Yo Intermittent Recovery test Level 1 distance (m) in a group of 119 highly trained soccer players who ranged in age from 11 to 18 years. The youth soccer players were subdivided into U11 ($n = 21$), U12 ($n = 14$), U13 ($n = 13$), U14 ($n = 16$), U15 ($n = 17$), U16 ($n = 14$), U17 ($n = 16$), and U18 ($n = 8$). All subjects participated in 2 testing sessions separated by 3–4 weeks.

During the first session, anthropometric measures were collected and a fitness battery including assessments of countermovement vertical jump with and without arm swing, 15-m sprint run, 15-m agility run, 20-m shuttle run, and the Yo-Yo intermittent recovery test was performed. The second test session was used to assess the players' repeated-sprint ability. Repeated-sprint ability

exhibited moderate to very large correlations with agility, moderate to large correlations with jumping ability, and small to moderate correlations with shuttle endurance tests. However, although these measures exhibit correlations, there is variability in these relationships between the age groupings. The most noted location for this uncoupling was between repeated-sprint ability and agility and explosive power. The highest correlations were found in the U18 group where repeated-sprint ability was correlated with acceleration, agility, and explosive leg power, which indicated a stabilization in the overall physical performance in this population. Based on these data, coaches should be mindful of the individual growth patterns during these age periods and consider that performance tests will exhibit greater variability with younger soccer players.

Spencer M, Pyne D, Santisteban J, and Mujika I. Fitness determinants of repeated-sprint ability in highly trained youth football players. *Int J Sports Physiol Perform* 6: 497–508, 2011.

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